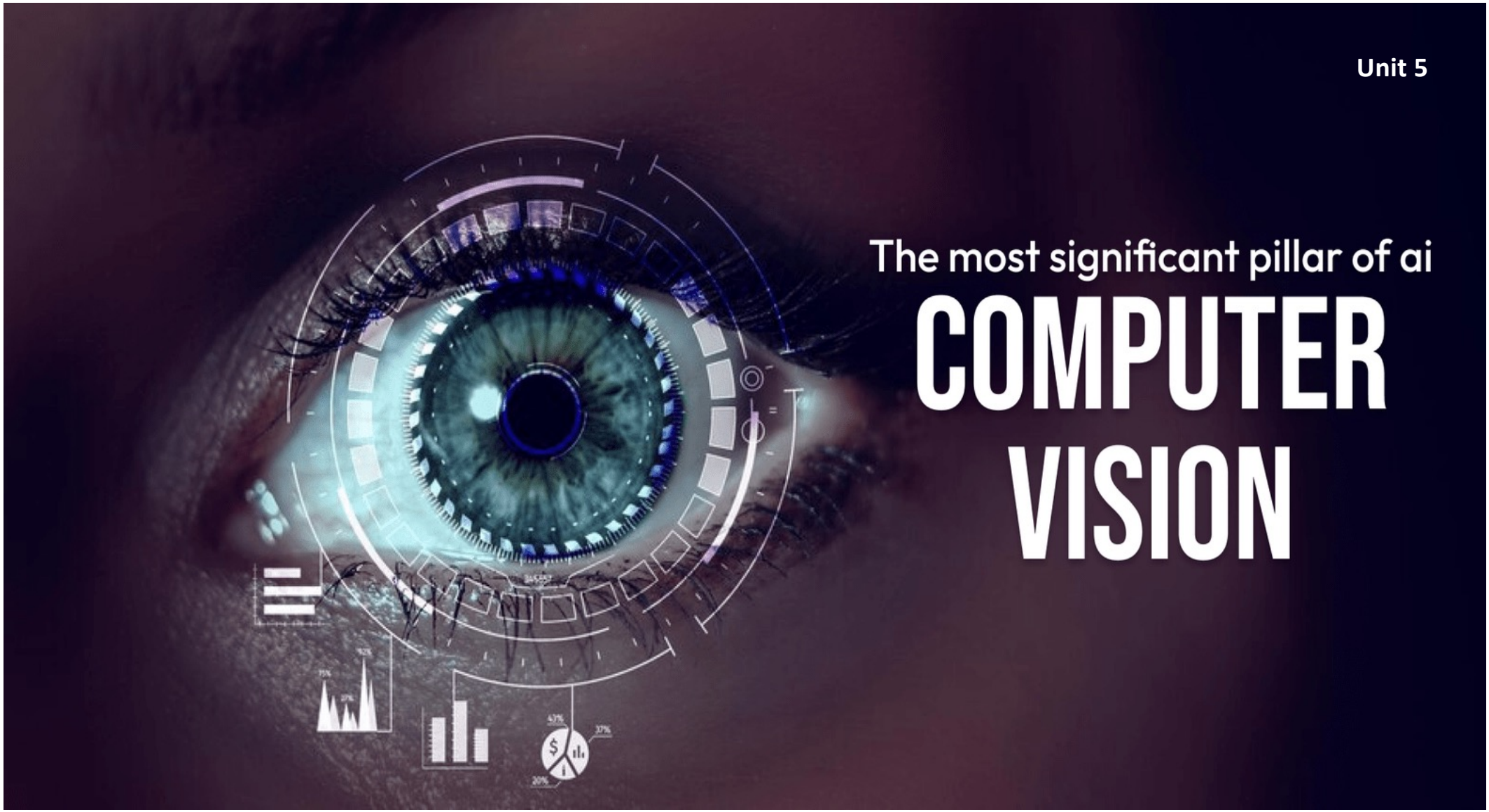
Unit 5

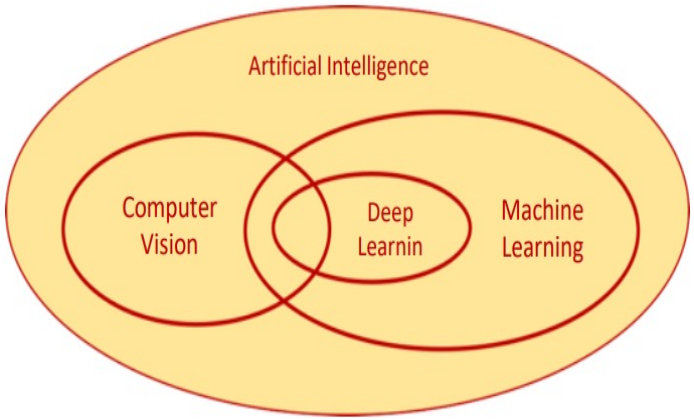


Overview

- Computer Vision is the process of extracting of information from images, text, videos, etc.,
- A system that can process, analyze and make sense of visual data in the same way as humans do.

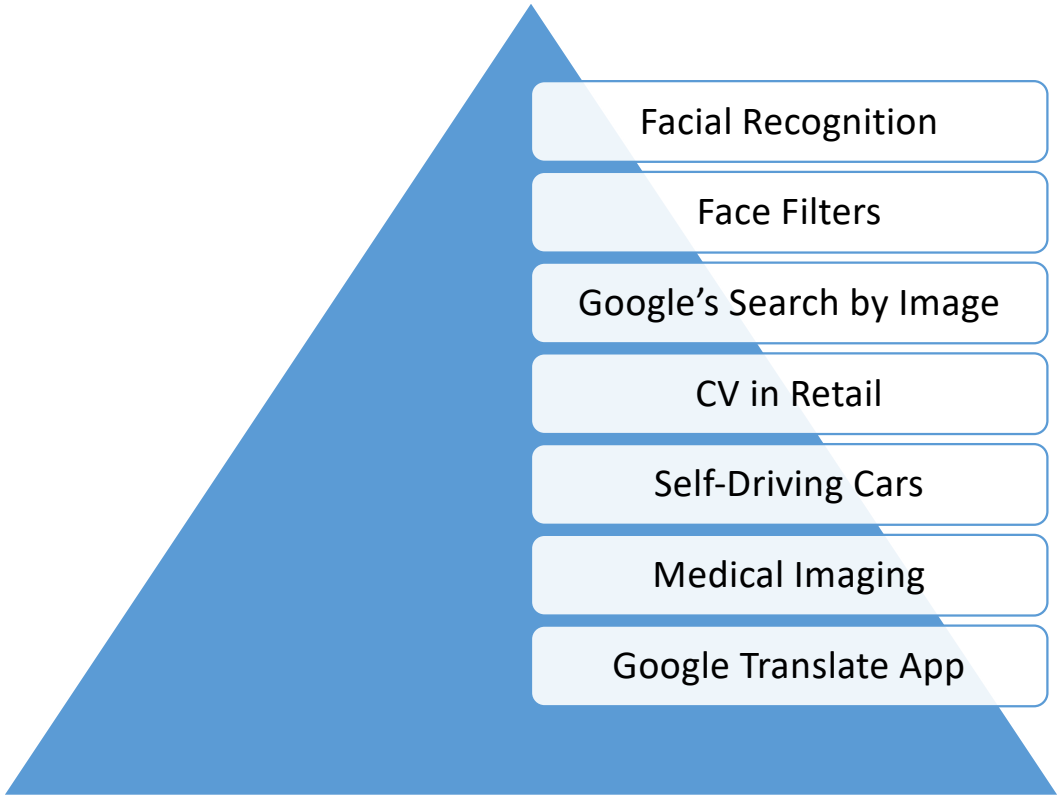
CV & AI

- Computer Vision is a field of AI.
- Computer Vision enables AI to see observe and make sense of visual data (such as images & videos)



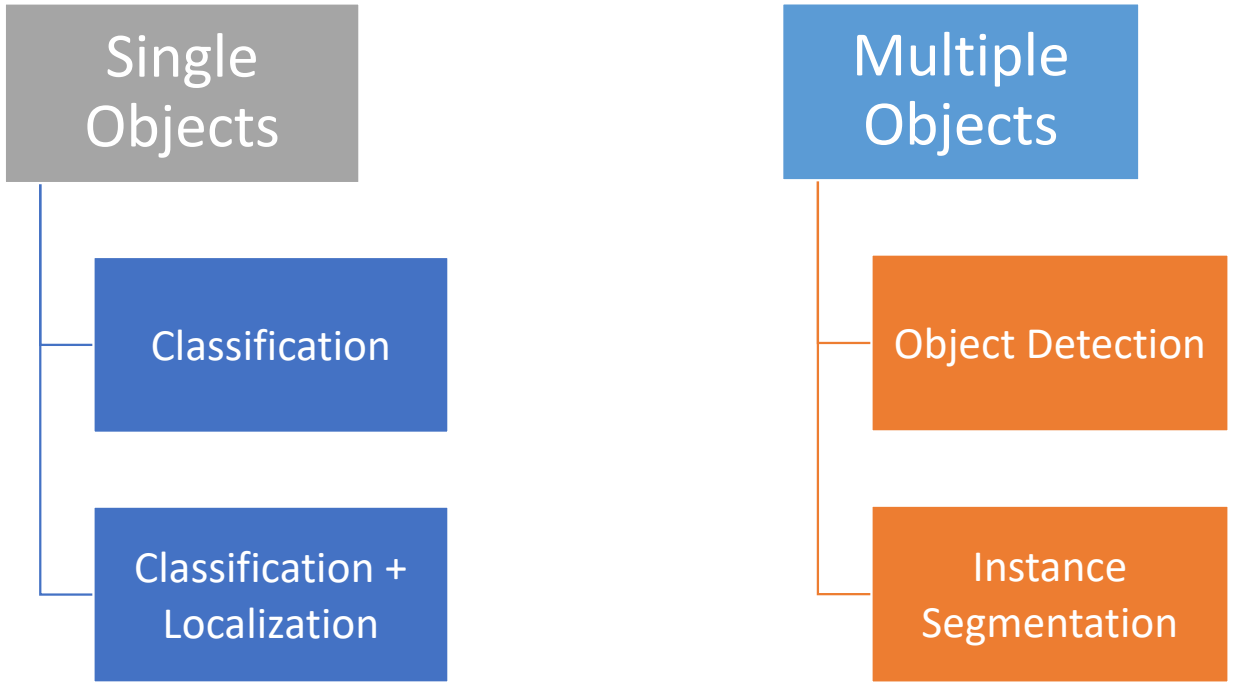
COMPUTER VISION – CV	IMAGE PROCESSING
○ Computer Vision involves extracting valuable information from images or videos, analyzing them to gain insights, and using that understanding to make predictions about the visual data.	▪ Image processing primarily focuses on manipulating raw images to improve their quality or prepare them for further tasks.
○ CV is a superset of image processing	▪ Image process is a subset of CV
○ Examples: Object Detection, Handwriting Recognition etc.,	▪ Example: Rescaling image, Correcting Brightness, Changing tones, etc.,

Applications of CV



CV tasks

Computer Vision rely on a set of tasks performed to extract specific information from the input image, which can either be used directly for prediction or serve as a foundation for further analysis.



1. Classification

Classification in CV refers to the tasks of identifying and categorizing objects, scenes, or patterns within an image into predefined categories or labels.

2. Classification+ Localization

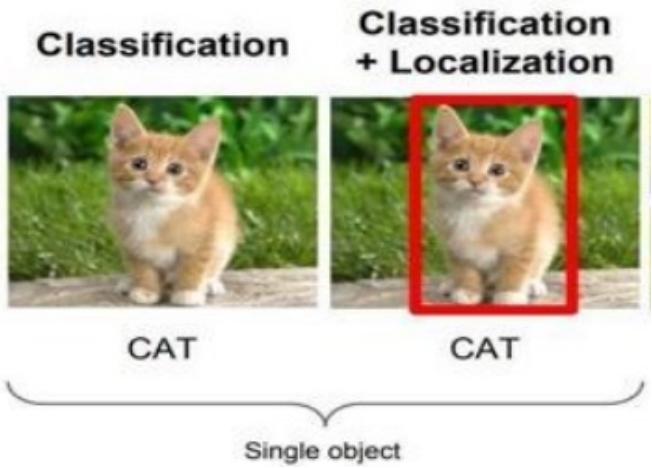
is a computer vision task where the model not only classifies what object is present in an image but also localizes it by drawing a bounding box around it.

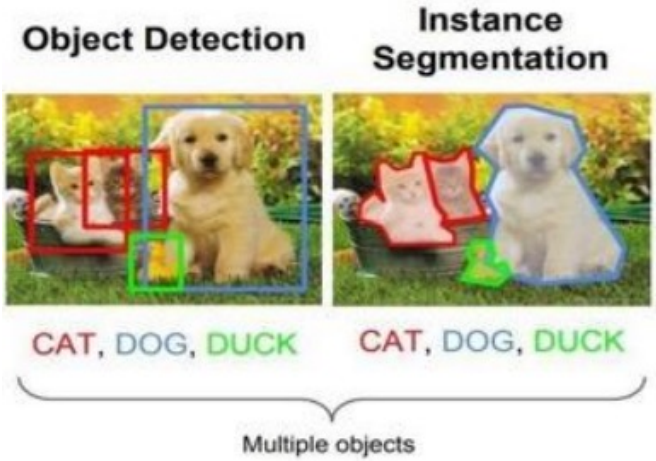
3. Object Detection

Is computer vision task that idenfies what objects are present in an images and where they are located - even when there are multiple objects.

4. Instance Segmentation

Instance segmentation involves detecting objects in an image, assigning them categories, and labeling each pixel that belongs to those objects. The output is a set of segments, each representing a specific object.





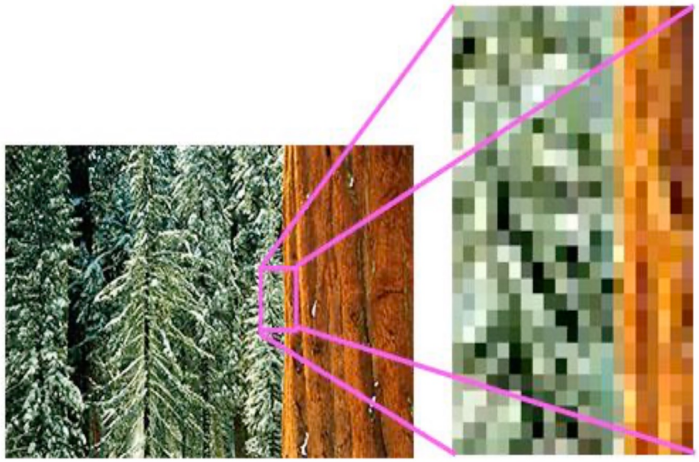
Basics of Images

We all see a lot of images around us and use them daily either through our mobile phones or computer system. But do we ask some basic questions to ourselves while we use them on such a regular basis.



Don’t know the answer yet? Don’t worry, in this section we will study about the basics of an image

Basics of Pixels



The word “pixel” means a picture element. Every photograph, in digital form, is made up of pixels. They are the smallest unit of information that make up a picture. Usually round or square, they are typically arranged in a 2-dimensional grid.

Resolution

Resolution refers to the number of pixels in an image. It can be expressed as width by height (e.g., 1280×1024), or as a single number (e.g., 5 megapixels, where one megapixel equals one million pixels). For example, a 1280×1024 monitor has about 1.31 megapixels (1280 x 1024 = 1,310,720 pixels).

Grey Scale Images

Each pixel in an image has a value that defines its brightness and color. In the common 8-bit format, this value ranges from 0 to 255, where 0 represents black and 255 represents white. This range comes from the binary system, where each pixel uses 1 byte (8 bits), and each bit can be either 0 or 1, allowing for 256 possible values (0 to 255).

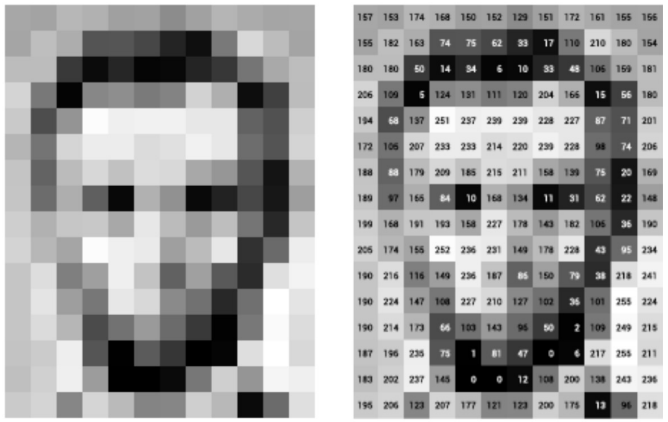
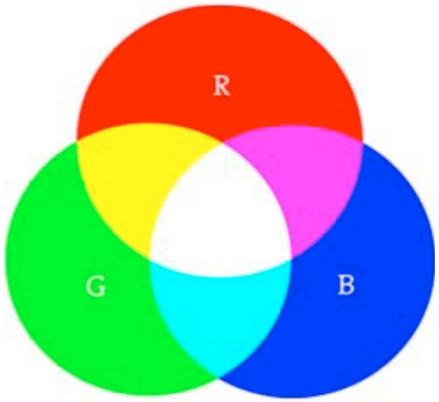


Image Features

In Computer Vision and image processing, a feature is a piece of information which is relevant for solving the computation tasks related to a certain application. Features may be specific structures in the image such as ***points, edges or objects***.

RGB Images

All the images that we see around are colored images. These images are made up of three primary colors Red, Green and Blue. All the colors that are present can be made by combining different intensities of red, green and blue.

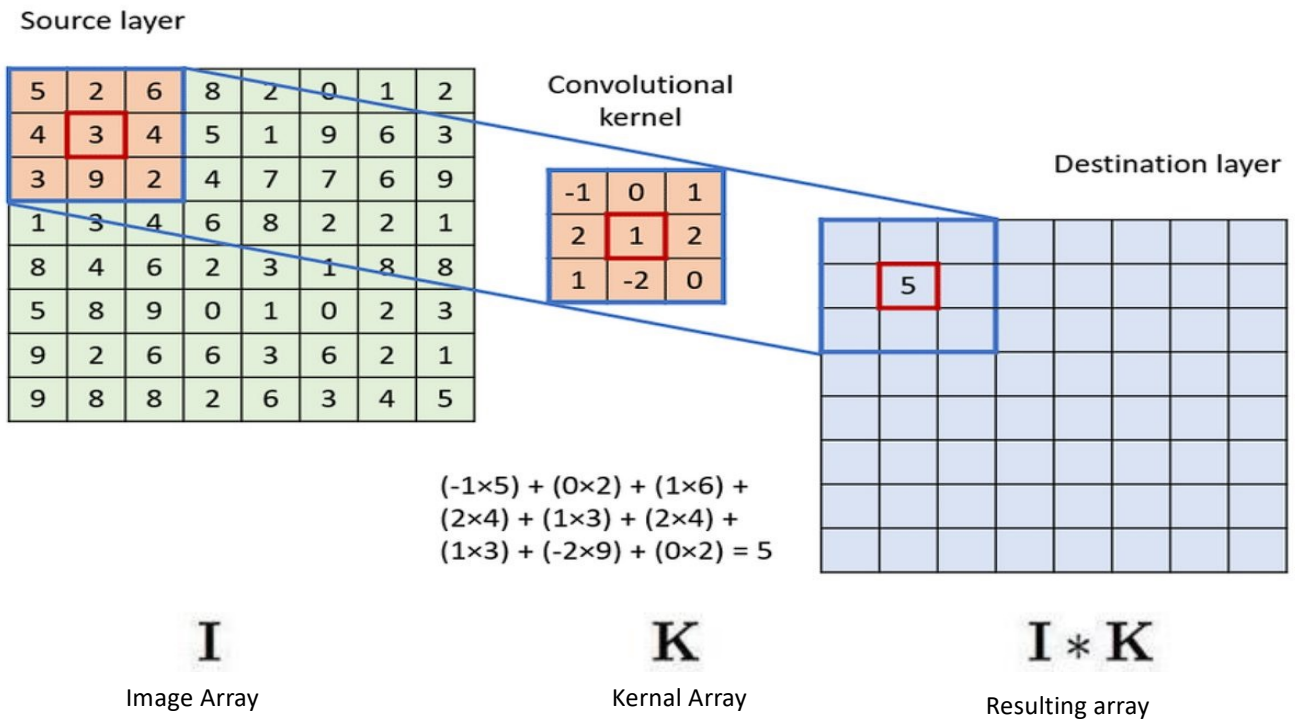


Task:
Go to the following link www.piskelapp.com and create your pixel art. Try and make a GIF using the online app for your pixel art.

Let us experience!
Go to this online link https://www.w3schools.com/colors/colors_rgb.asp.

What is Convolution?

Convolution in image processing is a mathematical operation that applies a small matrix, **called Kernel or Filter**, to an image to **create a new one by calculating a weighted sum of the pixels**.



What is Kernel?

In image processing, a **kernel, also known as a filter or mask**, is a small matrix of numbers used to apply effects like blurring, shapening, or edge detection to an image.

What is Convolution Neural Networks (CNN) ?

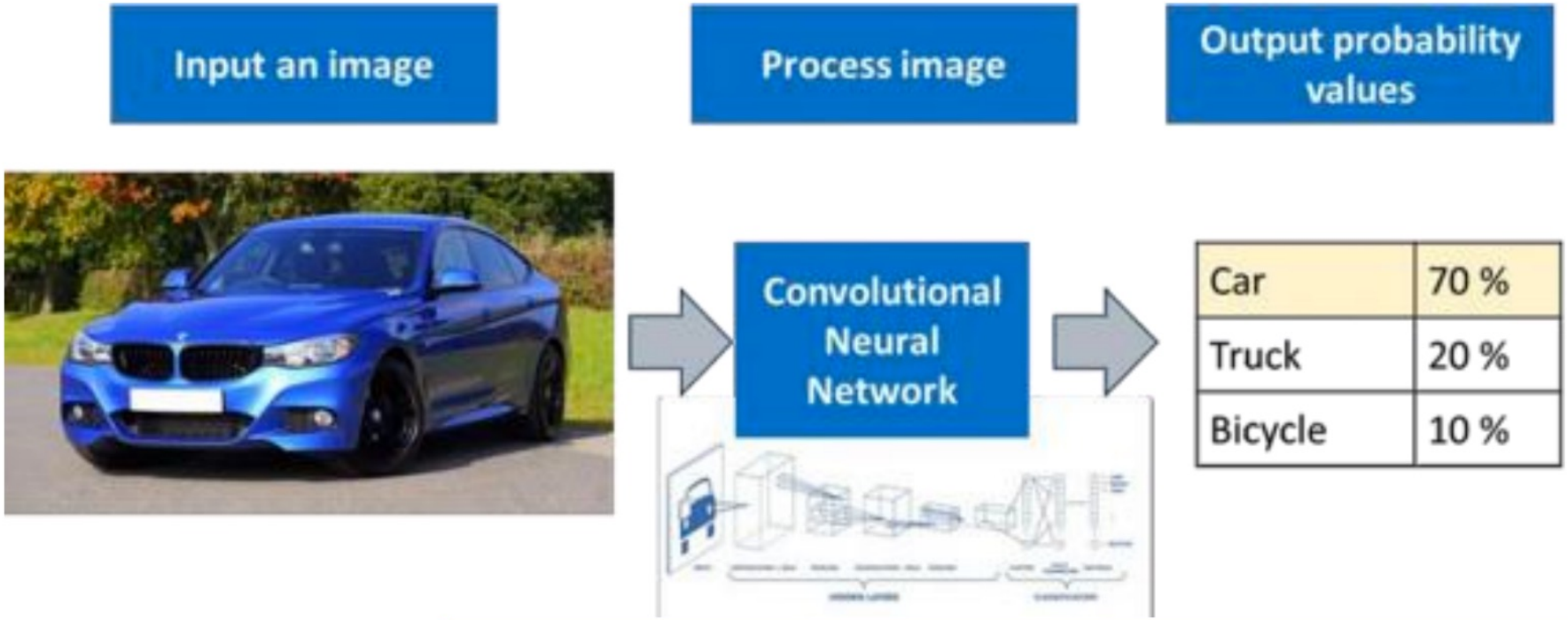
Introduction

A Convolutional Neural Network (CNN) is a type of deep learning model (algorithm) specifically designed for processing Image data with a grid-like structure, such as images.

In brief:

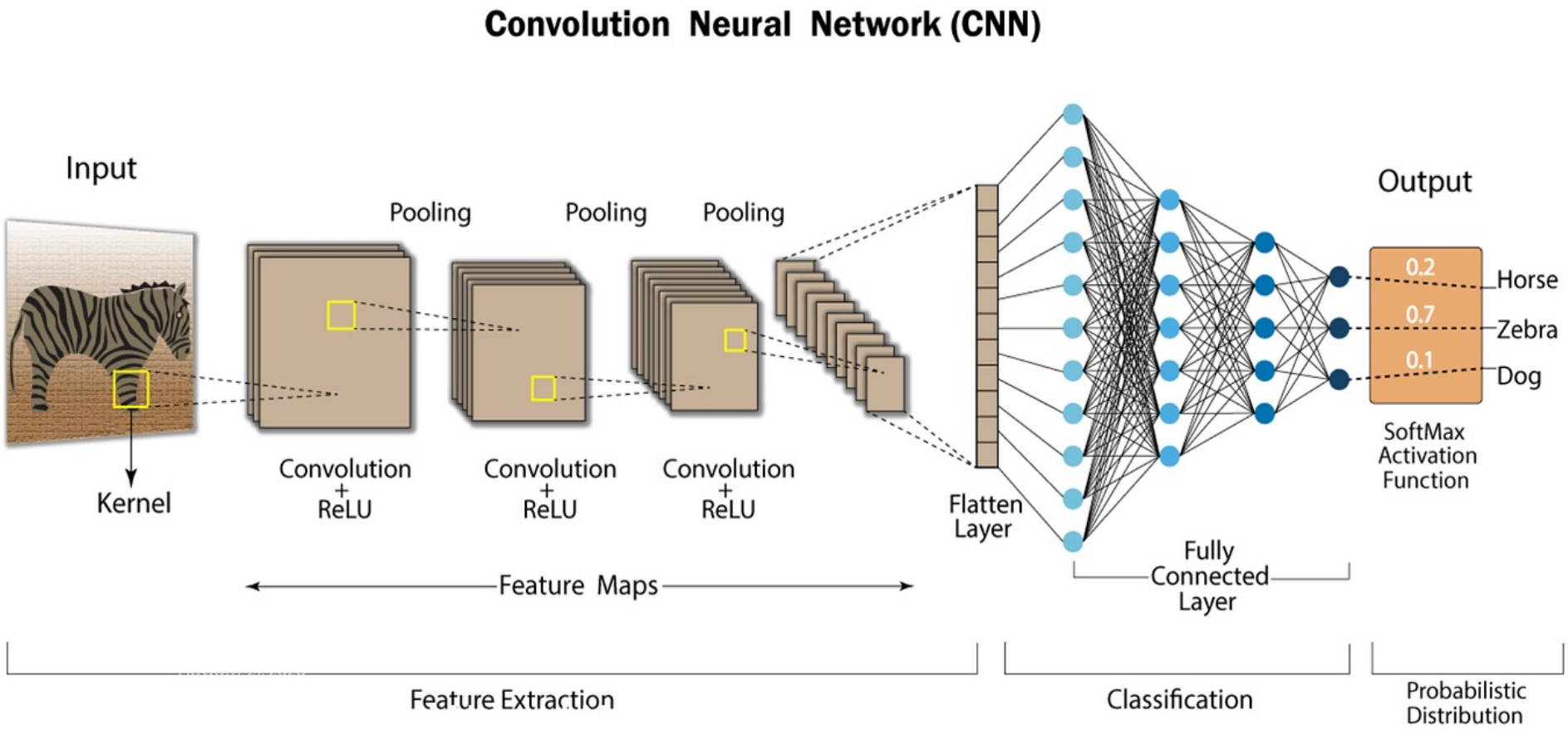
- ☐ CNNs use layers with convolution operations to automatically learn features from the input data.
- ☐ They are especially powerful for tasks like image classification, object detection, and image segmentation.
- ☐ Key components:
  - Convolutional layers: Extract features using filters/kernels.

The process of deploying a CNN is as follows:

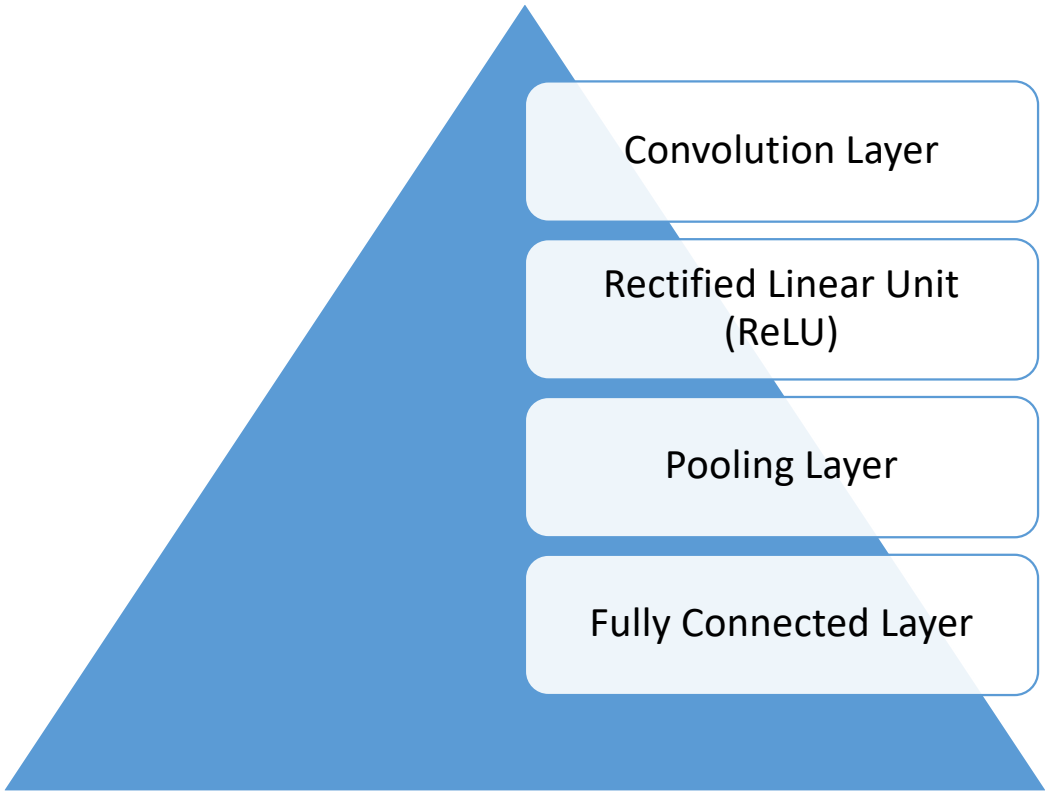


In the above diagram, we give an input image, which is then processed through a CNN and then gives prediction based on the label given in the particular dataset.

The different layers of a Convolutional Neural Network (CNN) are as follows:



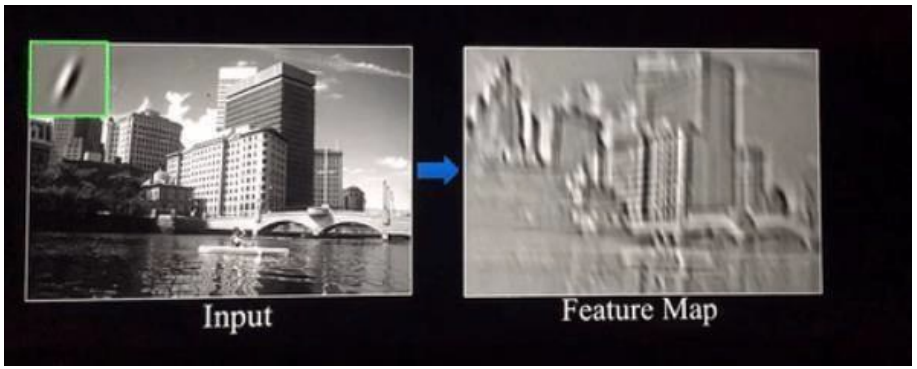
A Convolutional Neural Network of the following layers:



Convolution Layer

- The Objective of the Convolution Operation is to extract the high-level features such as **edges**, from the input images
- In the Convolution layer, several kernels are used to produce several features. The output of this layer is called the **feature map** also called an **activation map**.

1. It reduce the images size so that it can be processed more efficiently.
2. It focuses on the features of the image that can help in processing the image further

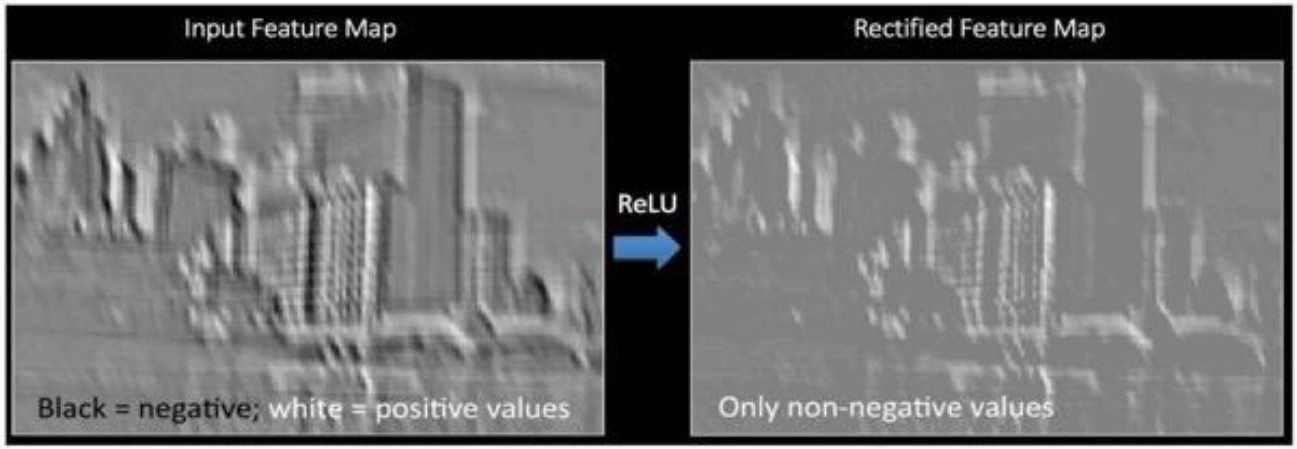


Rectified Linear Unit Layer - ReLU

- This layer simply gets rid of all the negative numbers in the features map and lets the positive number stay as it is.

In a CNN, the ReLU function:

- Introduces non-linearity so the network can learn complex patterns.
- Prevents vanishing gradients by keeping positive values unchanged and zeroing out negatives.



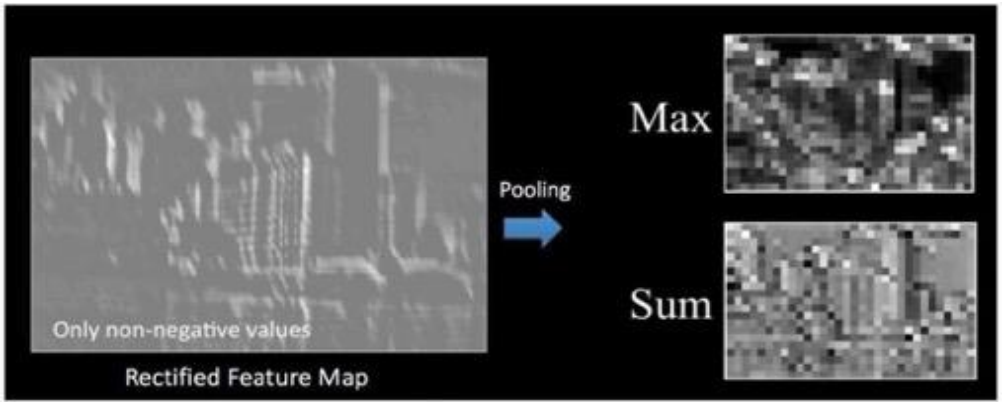
Pooling Layer

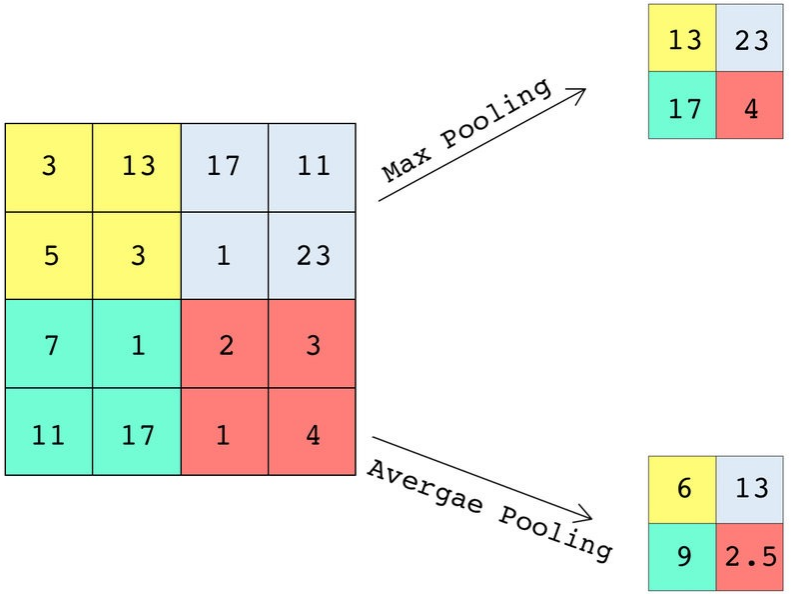
In short – Pooling helps summarize features and improve efficiency in CNN

- ❑ Downsamples the input (reduces dimensions)
- ❑ Reduces computation and memory usage.
- ❑ Makes features more robust to small shift or distortions.

Common types:

- ○ Max pooling: takes the maximum value in each region.
- ○ Average pooling: takes the average value in each region

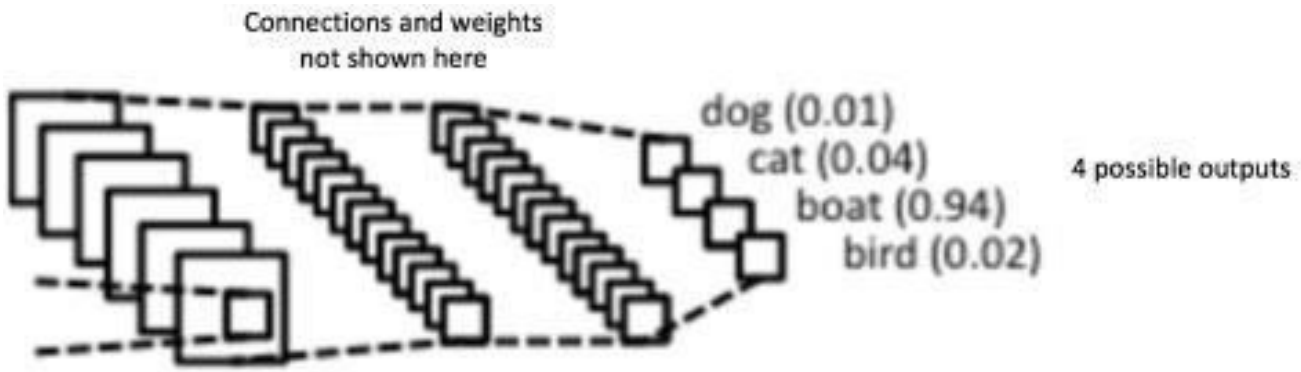




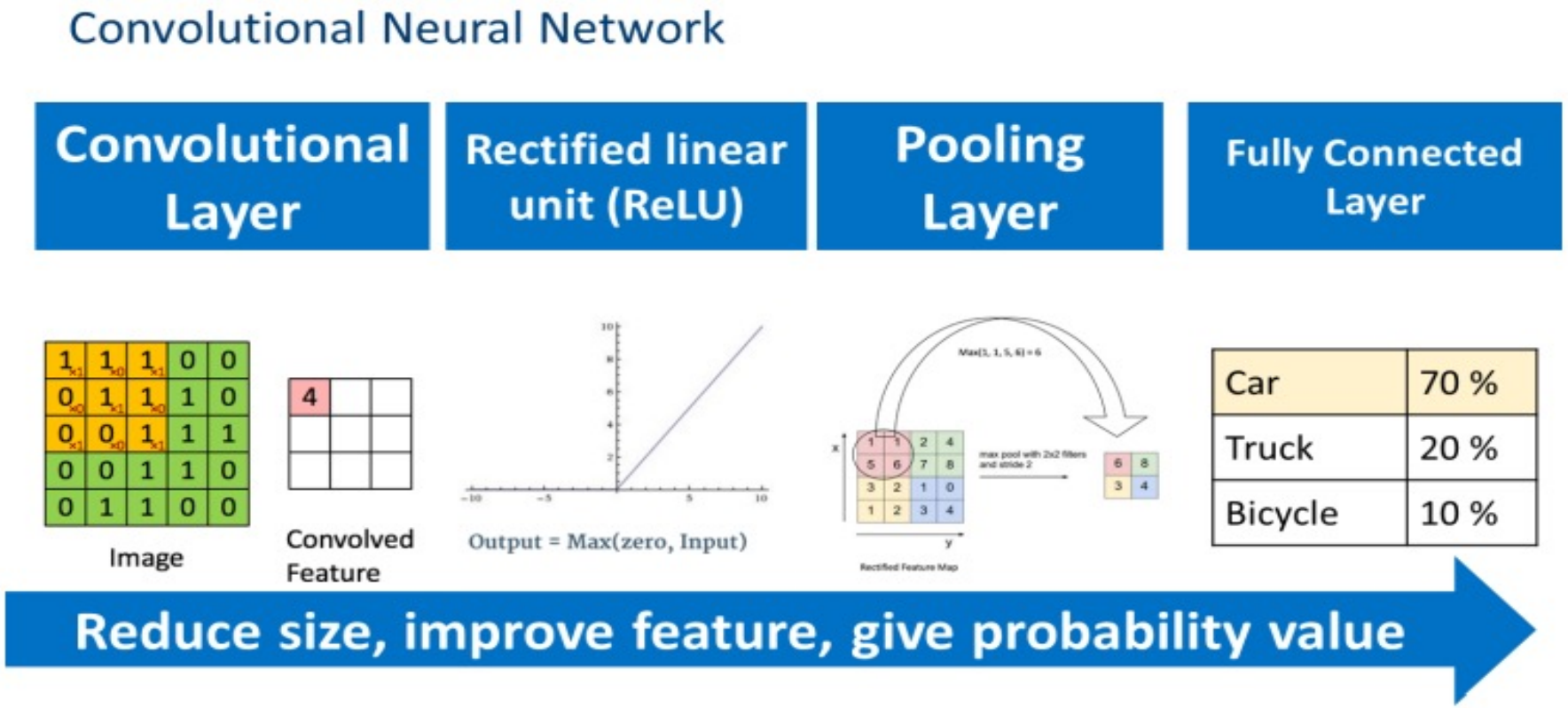
Fully Connected Layer

The objective of a fully connected layer is to make the results of the convolution/pooling process and use them to classify the image into a label (in a simple classification example).

- The output of convolution/pooling is flattened into a single vector of values, each representing a probability that a certain feature belongs to a label.
- For example, if the image is of a cat, features representing things like whiskers or fur should have high probabilities for the label “Cat”.



Let’s Summarize:




Thank
you!